

Corresponding parts of congruent triangles



- 1 Consider the statement,

$$\triangle JET \cong \triangle ZAP$$

State the angle that corresponds to the following:

- a $\angle EJT$ b $\angle JET$ c $\angle JTE$

- 2 Consider the statement,

$$\triangle ASK \cong \triangle TAX$$

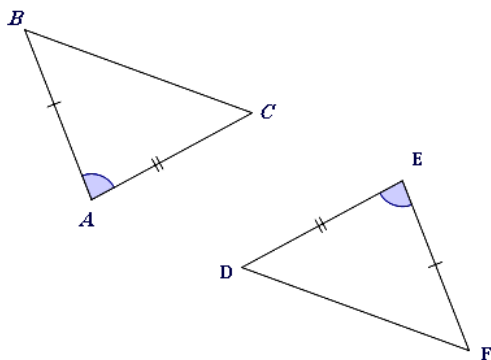
Determine whether the following can be concluded from the statement.

- a $\angle SKA \cong \angle ATX$ b $\angle AKS \cong \angle TXA$
c $\angle KAS \cong \angle XTA$ d $\angle ASK \cong \angle TAX$

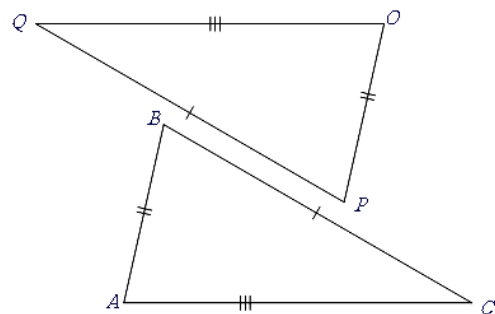
- 3 For each of the following diagrams:

- i State the congruence statement for the triangles.
ii State the reason why these two triangles are congruent.
iii Which angle is congruent to $\angle ACB$?

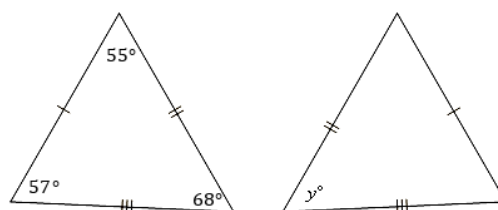
a



b



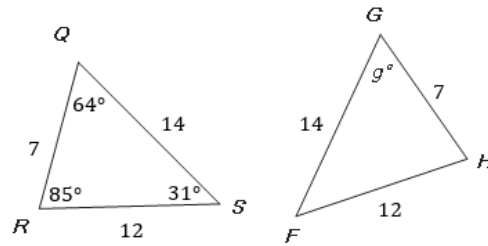
- 4 Consider the following diagram:
Find the value of y .



5 Consider the following pair of congruent triangles:



- a State the congruence statement for the triangles.
- b Identify the congruence postulate that justifies the congruence of the two triangles.
- c Hence, solve for g .

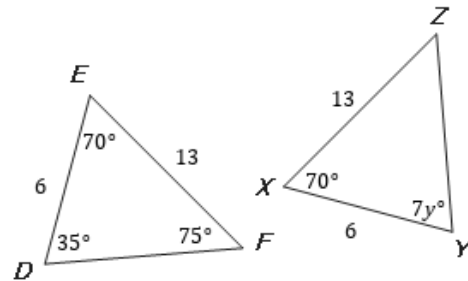


6 All corresponding sides and angles of $\triangle RST$ and $\triangle DEF$ are congruent. Determine whether the following statements are always true.

- a There is a sequence of transformations that maps $\triangle RST$ to $\triangle DEF$.
- b There is a translation followed by a rotation that maps $\overline{RT}$ to $\overline{DF}$.
- c There is a dilation that maps $\triangle RST$ to $\triangle DEF$.
- d There is a reflection that maps $\overline{RS}$ to $\overline{DE}$.
- e There is no sequence of rigid motions that maps $\triangle RST$ to $\triangle DEF$.

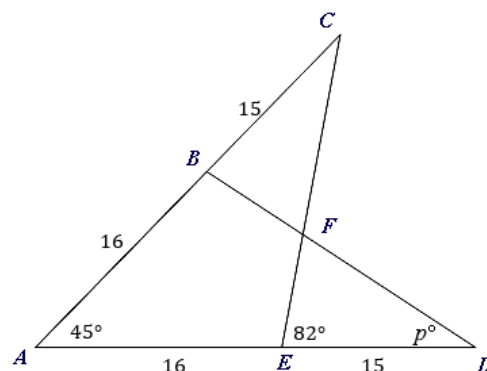
7 Consider the following diagram, where $m\angle XYZ = 7y^\circ$.

- a State the congruence statement for the triangles.
- b Identify the congruence postulate that justifies the congruence of the two triangles.
- c Hence, solve for y .

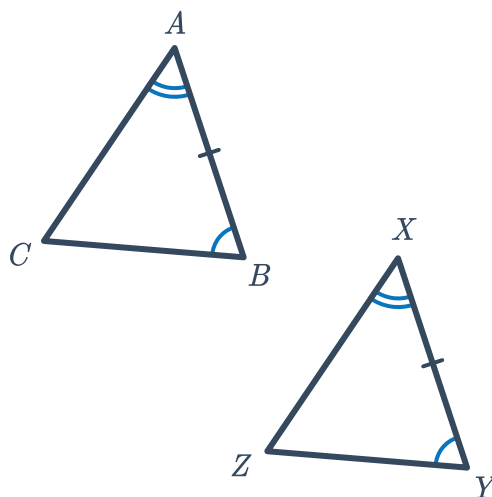


8 Consider the following figure:

- a Identify the congruence postulate that justifies $\triangle ACE \cong \triangle ADB$.
- b Hence, solve for p .



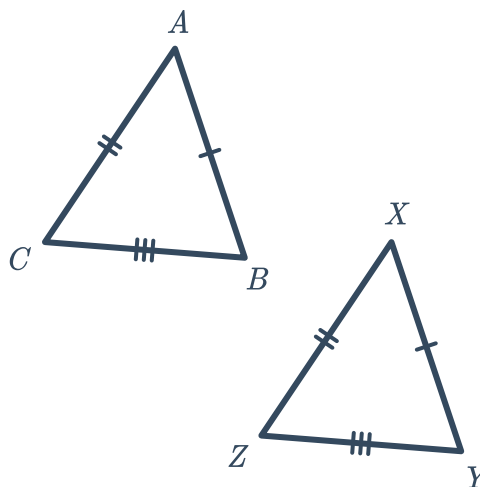
- 9 Use the information given in the diagram to complete the two-column proof that $\angle ACB \cong \angle XZY$.



To prove: $\angle ACB \cong \angle XZY$

Statements	Reasons
1. $\angle CAB \cong \angle ZXY$	Given
2. $\angle CBA \cong \angle ZYX$	Given
3. $\overline{AB} \cong \overline{XY}$	Given
4. $\triangle ABC \cong \triangle XYZ$	$\square$
5. $\angle ACB \cong \angle XZY$	Corresponding parts of congruent triangles are congruent (CPCTC)

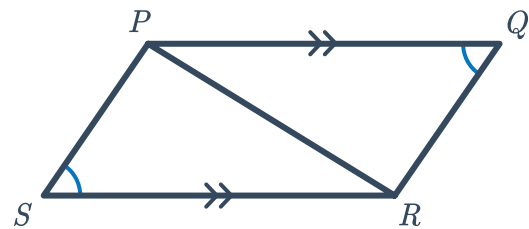
- 10 Use the information given in the diagram to complete the two-column proof that $\angle ACB \cong \angle XZY$.



To prove: $\angle \text{[Image]} \cong \angle XZY$

Statements	Reasons
1. $\overline{CB} \cong \overline{ZY}$	Given
2. $\overline{AC} \cong \overline{XZ}$	Given
3. $\overline{AB} \cong \overline{XY}$	Given
4. $\triangle ABC \cong \triangle XYZ$	Side-side-side congruence (SSS)
5. $\angle ACB \cong \angle XZY$	$\square$

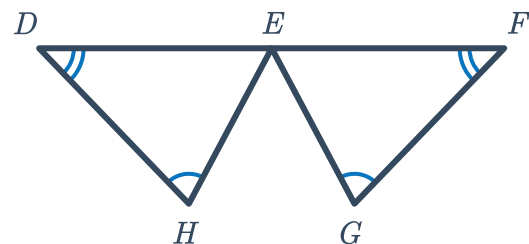
- 11 Use the information given in the diagram to complete the two-column proof that $\overline{SR} \cong \overline{PQ}$.



To prove: $\overline{SR} \cong \overline{PQ}$

Statements	Reasons
1. $\angle PSR \cong \angle RQP$	Given
2. $\overline{PQ} \parallel \overline{SR}$	Given
3. $\square$	$\square$
4. $\overline{PR} \cong \overline{RP}$	Reflexive property of congruence
5. $\triangle PQR \cong \triangle RSP$	Angle-angle-side congruence (AAS)
6. $\overline{SP} \cong \overline{RQ}$	$\square$

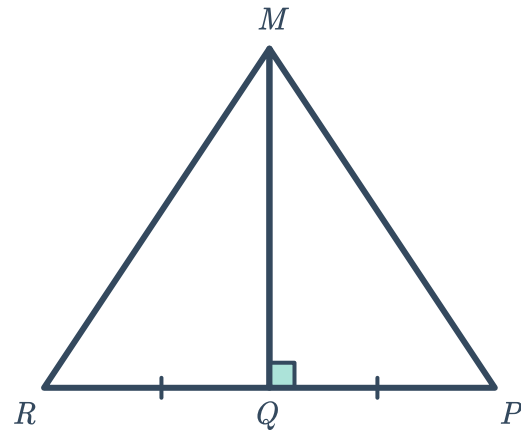
- 12 Use the information given in the diagram to complete the two-column proof that $\overline{DH} \cong \overline{FG}$.



To prove: $\overline{DH} \cong \overline{FG}$

Statements	Reasons
1. E is the midpoint of $\overline{DF}$	Given
2. $\angle EDH \cong \angle EFG$	Given
3. $\angle DHE \cong \angle FGE$	Given
4. $\square$	$\square$
5. $\triangle DEH \cong \triangle FEG$	Angle-angle-side congruence (AAS)
6. $\overline{DH} \cong \overline{FG}$	$\square$

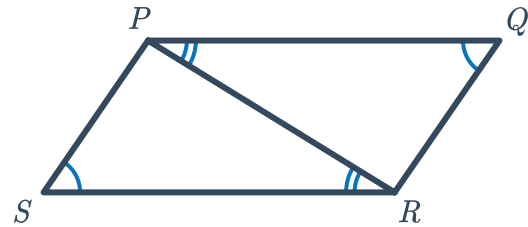
- 13 Use the information given in the diagram to complete the two-column proof that $\triangle RMP$ is isosceles.



To prove: $\triangle RMP$ is isosceles

Statements	Reasons
1. $\overline{RQ} \cong \overline{QP}$	Given
2. $\overline{RP} \perp \overline{MQ}$	Given
3. $\angle MQR$ and $\angle MQP$ are right angles	Definition of perpendicular
4. $\angle MQR \cong \angle MQP$	All right angles are congruent
5. $\overline{MQ} \cong \overline{MQ}$	Reflexive property of congruence
6. $\square$	$\square$
7. $\square$	$\square$
8. $\triangle RMP$ is isosceles	Isosceles triangles are triangles with two congruent sides

- 14 Consider the two-column proof which shows that $\overline{SP} \parallel \overline{QR}$ in the following diagram.



To prove: $\overline{SP} \parallel \overline{QR}$

Statements	Reasons
1. $\angle PSR \cong \angle RQP$	Given
2. $\angle SRP \cong \angle QPR$	Given

Determine whether the following correctly proves that $\overline{SP} \parallel \overline{QR}$, if added to the proof.

a

To prove: $\overline{SP} \parallel \overline{QR}$

Statements	Reasons
1. $\overline{PR} \cong \overline{PR}$	Reflexive property of congruence
2. $\triangle SRP \cong \triangle QPR$	Angle-angle-side congruence (AAS)
3. $\angle SPR \cong \angle QRP$	Corresponding parts of congruent triangles are congruent (CPCTC)
4. $\overline{SP} \parallel \overline{QR}$	Alternate interior angles converse

b

To prove: $\overline{SP} \parallel \overline{QR}$

Statements	Reasons
1. $\overline{SP} \parallel \overline{QR}$	Alternate interior angles converse

c

To prove: $\overline{SP} \parallel \overline{QR}$

Statements	Reasons
1. $\overline{PR} \cong \overline{PR}$	Reflexive property of congruence
2. $\triangle SRP \cong \triangle QPR$	Angle-angle-side congruence (AAS)
3. $\overline{SP} \parallel \overline{QR}$	Corresponding parts of congruent triangles are congruent (CPCTC)

d

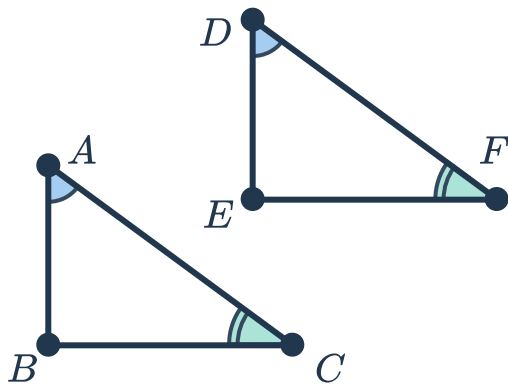
To prove: $\overline{SP} \parallel \overline{QR}$

Statements	Reasons
1. $\angle SPR \cong \angle QRP$	Third angles theorem
2. $\overline{SP} \parallel \overline{QR}$	Alternate interior angles converse

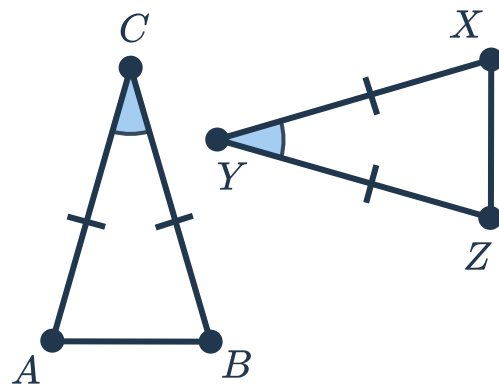
Additional questions

15 State the side corresponding to $\overline{AB}$ in each pair of congruent shapes:

a

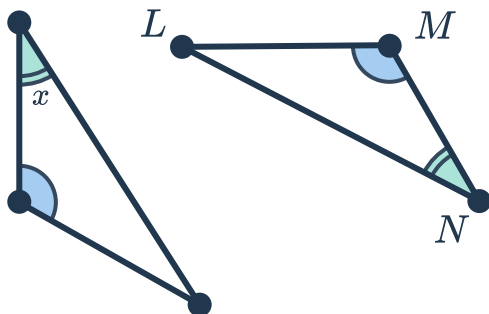


b

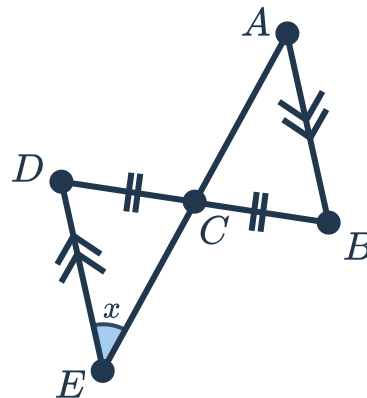


16 State the angle corresponding to the angle  a measure of x in each pair of congruent shapes:

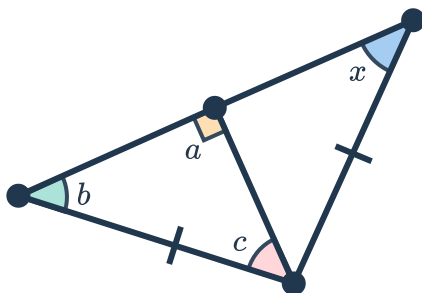
a



b



c



17 Consider the figure:

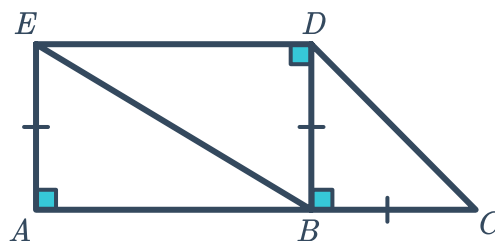
a Find the triangle congruent to $\triangle EAB$.

b Find the angle that corresponds to:

i $\angle EBA$ ii $\angle BEA$

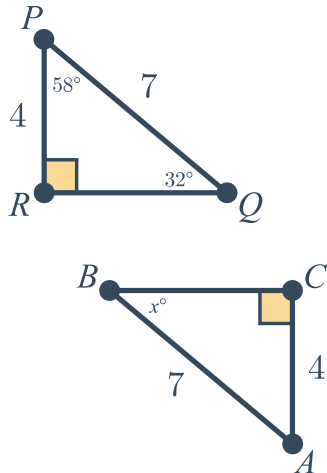
c Find the side that corresponds to:

i $\overline{AB}$ ii $\overline{AE}$

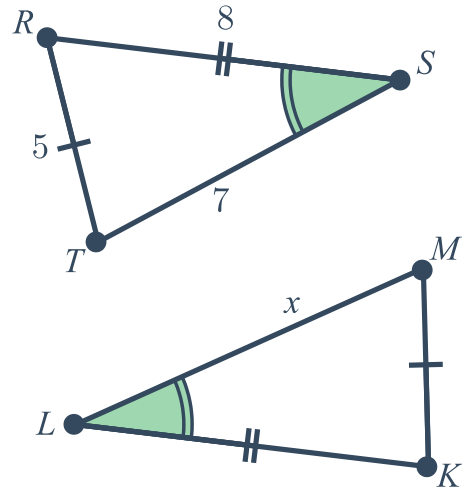


18 The triangles shown in the diagrams are congruent. Find the value of x .

a

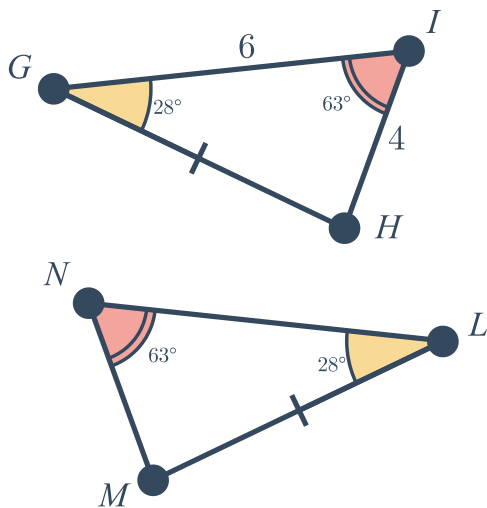


b

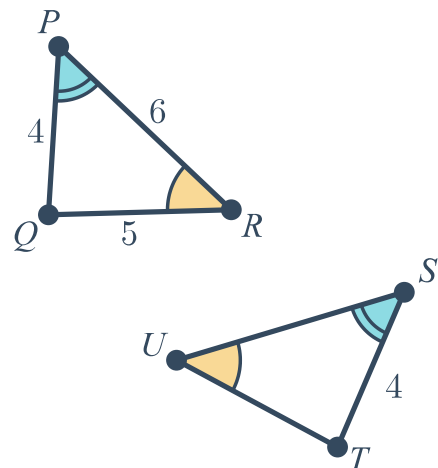


19 The triangles shown in the diagrams are congruent.

a Find NM .



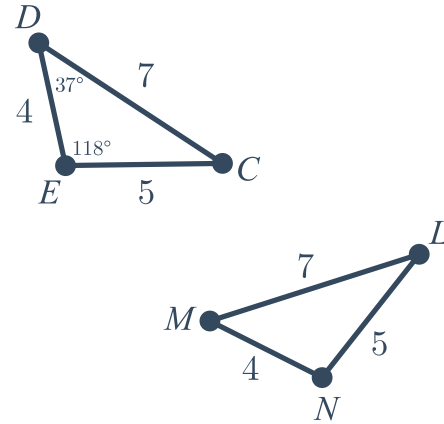
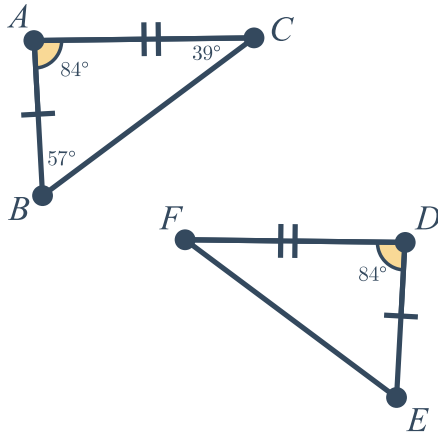
b Find SU .



c Find $m\angle DEF$.



Find $m\angle NLM$.



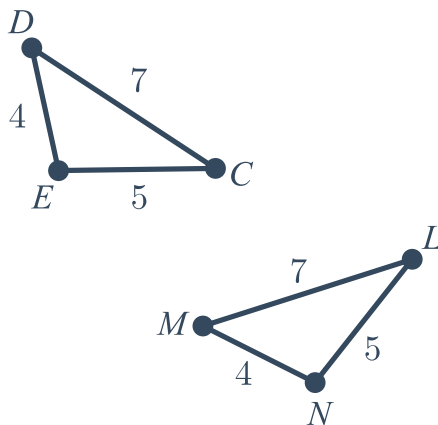
20

- Write two different congruence statements using the points D, E, F .
 $\triangle ABC \cong \triangle$ $\triangle ABC \cong \triangle$
- List all corresponding sides and angles for each congruence statement.
- Explain why the order of the points in the congruence statement is important.

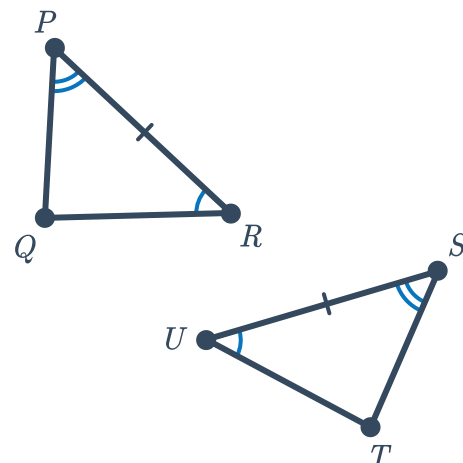
21

- Identify the congruence postulate justifies the congruence of the triangles shown in the diagram.
- Write a congruence statement for the triangles.
- Identify all additional pairs of congruent parts using the fact that corresponding parts of congruent triangles are congruent (CPCTC).

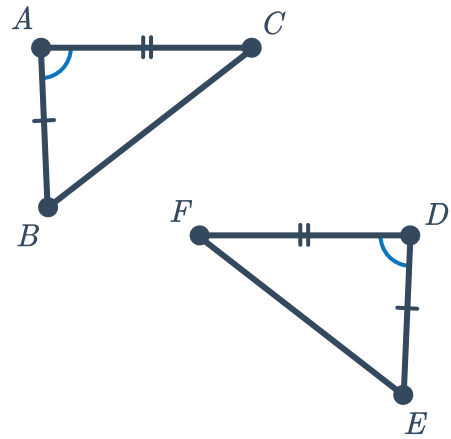
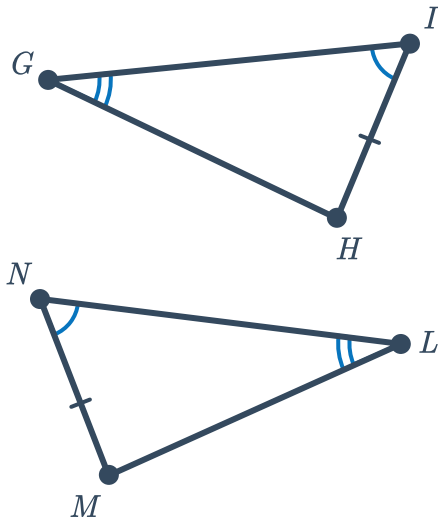
a



b



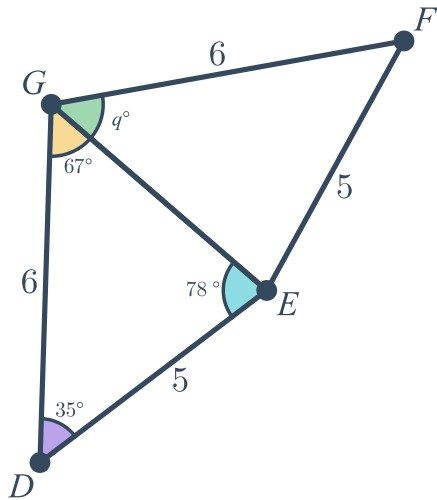
c



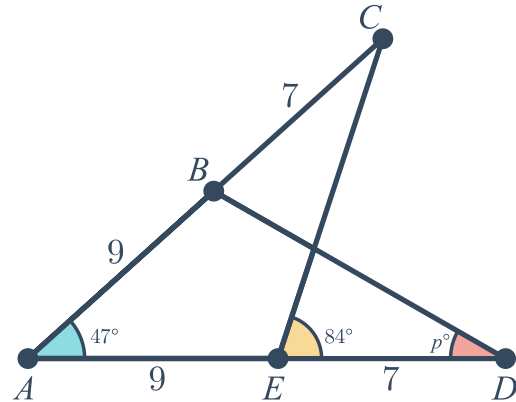
22 For each of the following diagrams:

- Identify the congruence postulate that justifies the congruence of the triangles shown in the diagram.
- Find the value of the variable.

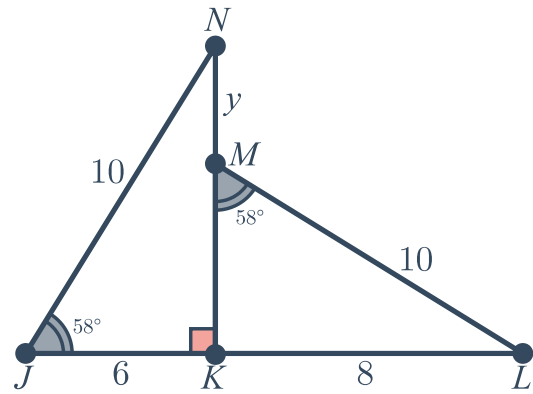
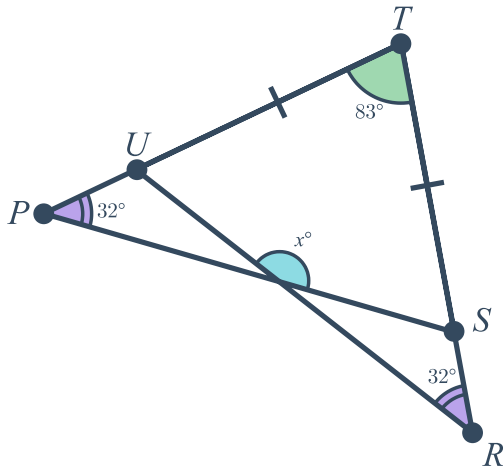
a



b



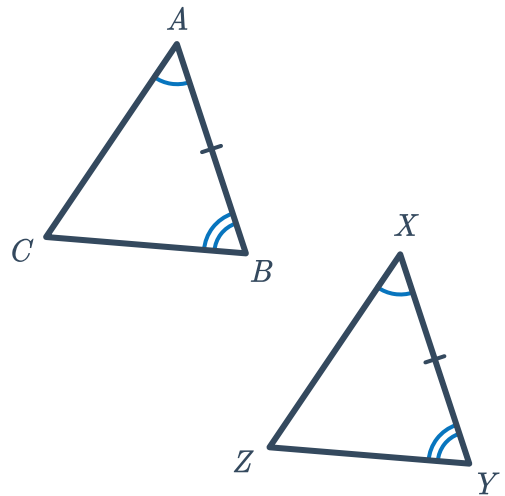
c



23 Given:

- $\angle A \cong \angle X$
- $\angle B \cong \angle Y$
- $\overline{AB} \cong \overline{XY}$

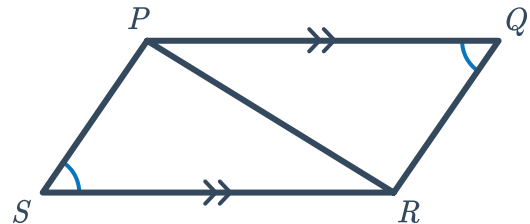
Prove: $\overline{BC} \cong \overline{YZ}$



24 Given:

- $\overline{PQ} \parallel \overline{RS}$
- $\angle S \cong \angle Q$

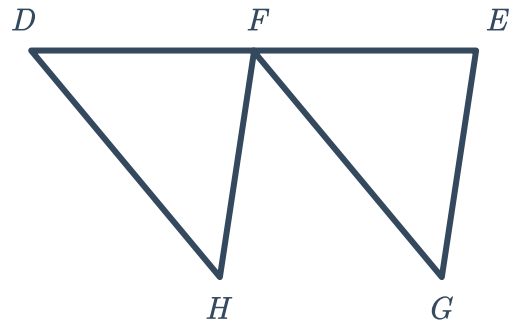
Prove: $\overline{PS} \cong \overline{QR}$



25 Given:

- F is the midpoint of $\overline{DE}$
- $\overline{DH} \parallel \overline{FG}$
- $\overline{DH} \cong \overline{FG}$

Prove: $\overline{FH} \parallel \overline{EG}$



26 Use the information given in the diagram to prove $\triangle PQX \cong \triangle RQX$.

